EE 800 Project Proposal

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Statement:

I pledge to adhere to the Stevens Graduate Student Code of Academic Integrity, and I have discussed the proposal with my Project Advisor.

Signiture:

Date: February 10, 2026

Vision-Based Navigation for Autonomous Mobile Robots in Persistent Monitoring Systems

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***Abstract*—This project develops a vision-based navigation framework for persistent monitoring using four-wheeled mobile robots operating in a dynamic environment. Persistent monitoring (PM) is the problem of continuously sensing a dynamic environment to detect changes immediately. This project addresses the primary challenge of implementing onboard visual perception for navigation under resource constraints. The system consists of a camera, a Raspberry Pi, a proximity sensor, and a four-wheeled mobile robot. The camera provides real-time input for target detection, while the Raspberry Pi processes the data and coordinates system control. The proximity sensor supports obstacle detection, and the robot executes movement commands, achieving autonomous navigation without external sensing infrastructure. Navigation between targets is achieved via an optimal tracking controller, and safety constraints are enforced to prevent collisions with obstacles. Closed-loop integration of perception, planning, and control allows the robot to continuously adapt to environmental changes.**

***Index Terms*—persistent monitoring, vision-based navigation, autonomous mobile robot, optimal tracking control**

I. Introduction

Persistent monitoring (PM) is the continuous monitoring of an environment, system, or set of targets over extended periods in order to detect changes promptly as they occur. Persistent monitoring has a wide range of applications, including environmental monitoring [1], infrastructure inspection [2], and security and surveillance [3]. In practical situations, the environment may change state, and random events may occur; hence, autonomous operation of the system is important.

Using robots for persistent monitoring presents many challenges. Robots in dynamic environments must balance conflicting objectives, minimizing travel time while ensuring safe operation [4]. They must also operate with limited computational power and sensing ability, especially on low-cost platforms. As a result, PM is essentially an optimization control problem in which robot trajectories are continuously replanned in response to environmental changes, while respecting kinematic and safety constraints.

Various navigation and control methods have been proposed for PM. Graph-based formulations are classic approaches, in which targets are represented as nodes and the robot solves variants of the traveling salesman or vehicle routing problems [5]. Some more recent methods rely on Receding Horizon Control (RHC), also known as Model Predictive Control (MPC) [6]. The controller uses a finite horizon to repeatedly update the target as new information becomes available. These methods provide strong theoretical approaches. However, many assume access to global position information from external infrastructure.

Vision-based navigation offers an ideal alternative for PM, particularly when external localization systems such as GPS are unavailable or unreliable [7]. Cameras can be considered low-cost sensors and provide ample information for persistent monitoring tasks. Vision ability allows the robot to directly acquire information about the environment and task, such as the appearance and spatial distribution of targets. In comparison, proximity sensors provide much less information and are only used to support obstacle detection in most cases [8].

Previous work on vision-based navigation includes classic feature-based methods and modern model-based methods. In classic target tracking problems, object detections are achieved with a closed-loop controller that adjusts the target position in the camera frame using PID [9]. For general navigation tasks, visual odometry and SLAM are used to provide pose estimates and local maps from camera data, whereas neural networks have been explored for processing visual data in complex environments [10]. In persistent monitoring problems, cameras are often integrated with receding-horizon planning to optimize routes online as targets move or new obstacles appear.

II. Problem Description

This project addresses the problem of vision-based persistent monitoring using an autonomous mobile robot operating in a dynamic environment. Persistent monitoring is defined as the repeated observation of a set of predefined targets, whose states may change over time, while minimizing travel time between targets and maintaining safe operation.

The system is implemented on a four-wheeled mobile robot (ELEGOO Smart Robot Car Kit V4.0) equipped with a camera, a proximity sensor, and an embedded processor, as shown in Fig. 1. The robot operates without external localization or sensing infrastructure and only relies on the onboard perception and computation module.

Formally, given (i) a bounded flat environment with obstacles, (ii) a set of targets with known locations, and (iii) a mobile robot subject to kinematic and actuation constraints, the problem is to design a closed-loop navigation system that:

Fig. 1. Photograph of the Elegoo robot used in this project.

1) Performs visual perception, path planning, and control computation with the Raspberry Pi.
2) Uses a camera to localize targets and detect environmental changes in real time.
3) Generates and updates paths online using a receding horizon strategy to ensure continuous target visitation.
4) Executes planned paths while enforcing safety constraints that avoid collision with obstacles.
5) Translates control commands into pulse-width modulation (PWM) signals via the embedded processor to drive the wheel motors.

The block diagram of the overall system is shown in Fig. 2.

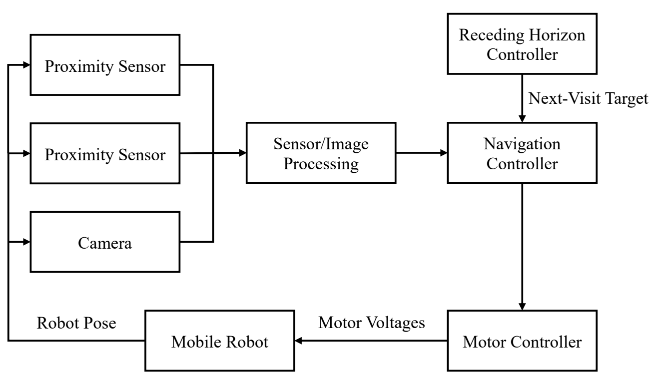

Fig. 2. Block diagram of the system

III. Expected Contributions

- A vision-based persistent monitoring framework that integrates onboard perception, planning, and control without reliance on external localization or sensing infrastructure.
- A low-cost system implementation using a camera, Raspberry Pi, and a four-wheeled mobile robot, demonstrating practical feasibility.
- Experimental validation of the proposed framework showing adaptive behavior, robustness to environmental changes, and reliable persistent monitoring performance.
- Open-source release of algorithms and system implementation to support reproducibility and further research.

IV. Project Timeline

The project is scheduled to conclude in four months and is divided into three phases aligned with the system's design, implementation, and validation, as shown in Fig. 3.

Phase 1: (Early February) Phase 1 focuses on proposal and system design. The system requirements and objectives are defined, and the overall architecture is explored.

Phase 2: (February to March) Phase 2 concentrates on system development and integration, including the assembly of hardware and development of algorithms for image processing, navigation, and motor control.

Phase 3: (April to May) Phase 3 is focused on experimentation and system validation. The robot will be tested in a dynamic environment to evaluate its performance. System parameters will be tuned based on experimental outcomes, followed by a detailed analysis of the results.

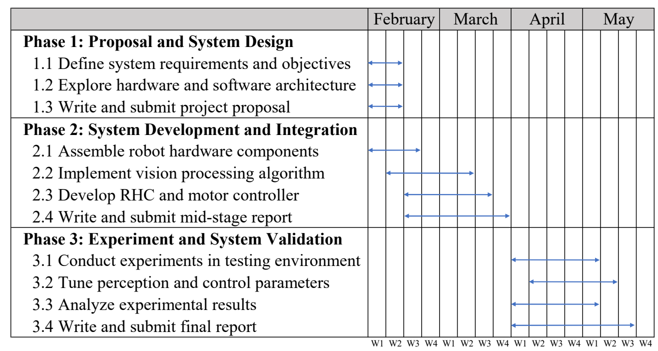

Fig. 3. Estimated timeline of this project.

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